

# Invertible functions and inverse functions



# Outline

Invertible functions and inverse functions

Graph of the inverse function

Inverse power functions

Inverse trigonometric functions

Inverse hyperbolic functions

# Invertible functions and inverse functions



# Introduction: Invertible functions and inverse functions

- ▶ A function associates to each input  $x$  a unique output  $y = f(x)$ : we ask when it is possible to “go back”, that is, to recover from  $y$  the unique  $x$  that produces it
- ▶ In this section we define **invertible functions** and the **inverse function**, with a graphical criterion to recognize them
- ▶ We will prove that **strictly monotone** functions are invertible, and we will see that this is not a necessary condition

# Invertible functions

- ▶ Given a real function of a real variable  $f : D \rightarrow \mathbb{R}$ , for every input  $x$  in the domain  $D$  there exists a unique output  $y = f(x)$  in the image of the domain  $f(D)$ .
- ▶ If, in addition, for every output  $y = f(x) \in f(D)$  there exists a unique input  $x \in D$ , then  $f$  is called **invertible**, and it establishes a **one-to-one correspondence** between the domain  $D$  of  $f$  and the image of the domain  $f(D)$ .

## Definition 1: of invertible function

A function  $f : D \rightarrow \mathbb{R}$  is **invertible** on the domain  $D$  if one of the following equivalent conditions holds:

$$\forall x_1, x_2 \in D, \quad x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$$

$$\forall x_1, x_2 \in D, \quad f(x_1) = f(x_2) \implies x_1 = x_2$$

$$\forall y \in f(D), \quad \exists! x \in D : y = f(x)$$

that is, if  $f$  is **injective**.

# Warning

- ▶ **WARNING:** the definition of invertible function given here for real functions of a real variable is different from the usual one given for functions between arbitrary sets, which also requires surjectivity.
- ▶ In this case *injectivity is enough* because we define the inverse on the image of  $f$  and not on its whole codomain.

## Definition 2: of inverse function

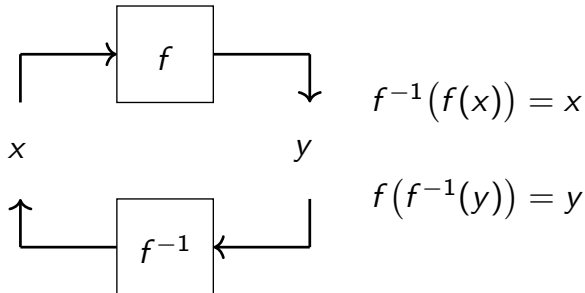
*Given an invertible function  $f : D \rightarrow \mathbb{R}$ , the function that associates to each output  $y \in f(D)$  the unique input  $x \in D$  such that  $f(x) = y$  is called the **inverse function** of  $f$  and is denoted by the symbol  $f^{-1}$ .*

## The pair $f$ and $f^{-1}$

The pair  $f$  and  $f^{-1}$  is written as:

$$f : \begin{cases} y = f(x) \\ x \in D \end{cases} \quad f^{-1} : \begin{cases} x = f^{-1}(y) \\ y \in f(D) \end{cases}$$

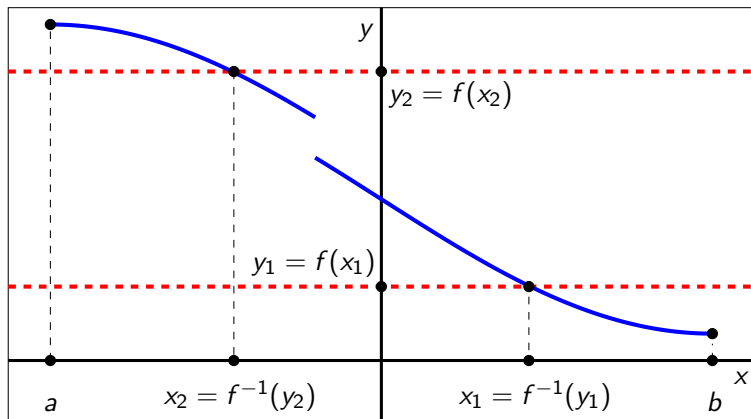
The black box of  $f^{-1}$  works backwards with respect to that of  $f$ , according to the following scheme:



# Graphical criterion for invertibility

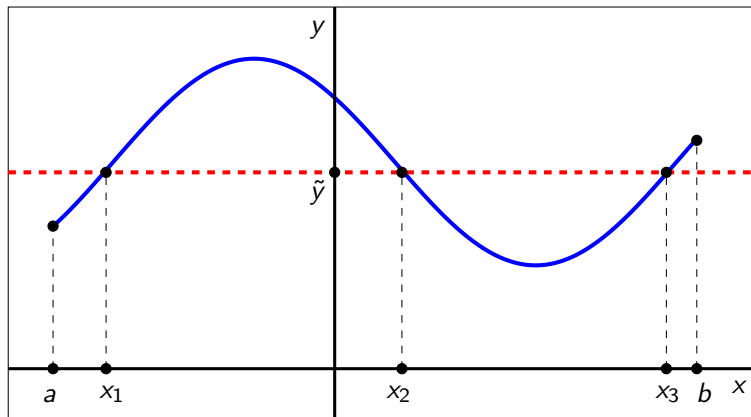
- ▶ The invertibility condition is equivalent to requiring that the graph of  $f$  be intersected in at most one point by every line parallel to the  $x$ -axis.
- ▶ We now see an example of the graph of an invertible function and one of a non-invertible function

## Example



Graph of a function that is **invertible** on  $[a, b]$ , since every line parallel to the  $x$ -axis either does not intersect the graph of  $f$  or intersects it in exactly one point.

## Example



Graph of a function that is **not invertible** on  $[a, b]$ , since for the indicated value  $\tilde{y}$  there are several values  $x$  (precisely  $x_1$ ,  $x_2$  and  $x_3$ ) whose image is  $\tilde{y}$ .

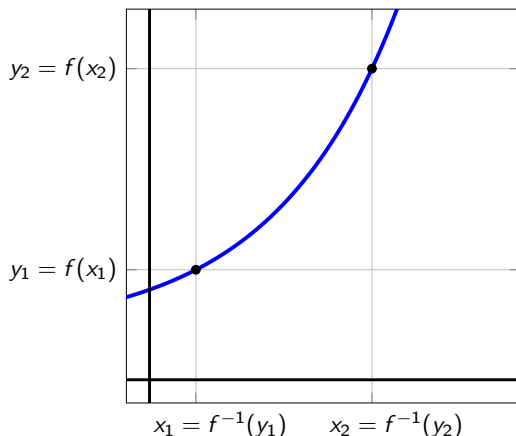
# Strictly monotone functions

## Theorem 1

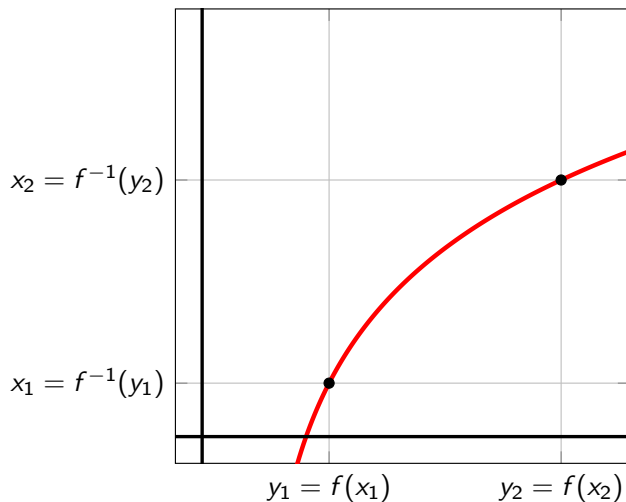
*If a function  $f : D \rightarrow \mathbb{R}$  is strictly increasing (decreasing) on  $D$ , then it is invertible on  $D$ . Moreover, its inverse function is strictly increasing (decreasing).*

# Graphical interpretation (1/2)

For the strictly increasing case, graphically we have:



## Graphical interpretation (2/2)



## Proof (1/2)

Consider the case of a function that is strictly increasing on  $D$  and two arbitrary values  $x_1, x_2 \in D$  (the strictly decreasing case is analogous).

If  $x_1 \neq x_2$ , then either  $x_1 < x_2$  or  $x_1 > x_2$ . Since  $f$  is strictly increasing, we have:

$$\text{if } x_1 < x_2, \text{ then } f(x_1) < f(x_2)$$

$$\text{if } x_1 > x_2, \text{ then } f(x_1) > f(x_2)$$

In both cases  $f(x_1) \neq f(x_2)$ , therefore  $f$  is invertible.

## Proof (2/2)

Consider the inverse function  $f^{-1}$ , that is,  $x = f^{-1}(y)$ , and let us prove that it is strictly increasing. Consider two arbitrary values  $y_1, y_2 \in f(D)$ , with  $y_1 < y_2$ .

If we had  $f^{-1}(y_1) = x_1 \geq x_2 = f^{-1}(y_2)$ , since  $f$  is strictly increasing we would have  $y_1 = f(x_1) \geq f(x_2) = y_2$ , that is,  $y_1 \geq y_2$ . This cannot happen, since it contradicts  $y_1 < y_2$  (**contradiction**).

Consequently, we have  $f^{-1}(y_1) = x_1 < x_2 = f^{-1}(y_2)$ , that is,  $f^{-1}(y_1) < f^{-1}(y_2)$ , and hence  $f^{-1}$  is strictly increasing. □

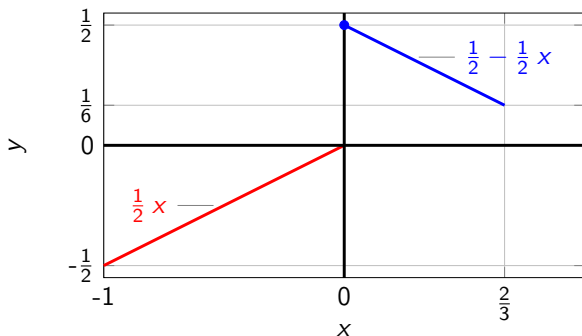
# Strict monotonicity is not necessary

- ▶ However, a function can be invertible even without being strictly increasing or decreasing.

## Example

For example, the following piecewise-defined function is invertible:

$$f(x) = \begin{cases} \frac{1}{2}x & \text{if } -1 \leq x < 0, \\ \frac{1}{2} - \frac{1}{2}x & \text{if } 0 \leq x \leq \frac{2}{3} \end{cases}$$



# Non-invertible functions

- ▶ Two classes of functions that are certainly not invertible are:
  1. even (symmetric) functions, since:

$$f(-x) = f(x)$$

2. periodic functions, since:

$$f(x + T) = f(x)$$

# Summary

## Invertible and inverse functions:

- ▶  $f$  is invertible on  $D$  if it is injective: equal outputs correspond to equal inputs
- ▶ The inverse  $f^{-1}$  is defined on the image  $f(D)$  and associates to each  $y$  the unique  $x$  with  $f(x) = y$
- ▶ Graphical criterion: every horizontal line intersects the graph in at most one point

## Monotonicity:

- ▶ Strictly monotone  $\Rightarrow$  invertible, with inverse monotone of the same type (the converse does not hold)
- ▶ Even functions and periodic functions are not invertible

# Graph of the inverse function

# Introduction: Graph of the inverse function

- ▶ In this section we see how to obtain the **graph** of  $f^{-1}$  from that of  $f$ : a symmetry with respect to the bisector  $y = x$  is enough
- ▶ We then see how to find the **analytic expression** of the inverse by solving the equation  $f(x) = y$  for  $x$
- ▶ Examples: affine function,  $x^2$ , exponential; and a case in which the inverse exists but cannot be written explicitly

## Symmetry with respect to the bisector

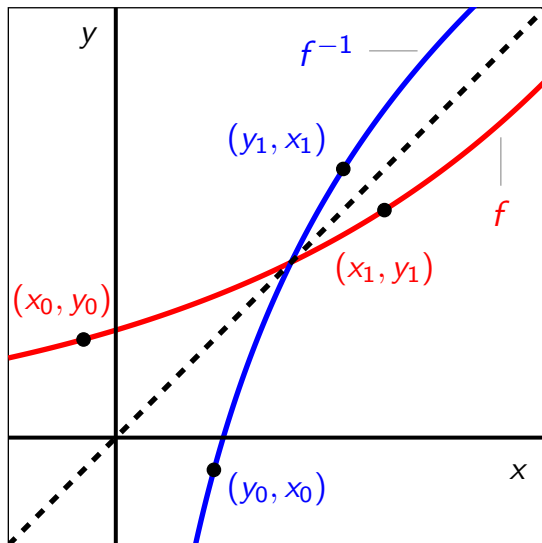
- ▶ The relations between an invertible function  $f$  and its inverse function  $f^{-1}$ :

$$f : \begin{cases} y = f(x) \\ x \in D \end{cases} \qquad f^{-1} : \begin{cases} x = f^{-1}(y) \\ y \in f(D) \end{cases}$$

indicate that if the point  $(x_0, y_0)$  lies on the graph of  $f$ , then the point  $(y_0, x_0)$  lies on the graph of  $f^{-1}$ .

- ▶ Since the points  $(x_0, y_0)$  and  $(y_0, x_0)$  are **symmetric with respect to the bisector** with equation  $y = x$ , the graph of  $f^{-1}$  is obtained from that of  $f$  by symmetry with respect to the bisector.

## Symmetry with respect to the bisector



# Analytic expression of the inverse

- ▶ If the analytic expression of  $f$  is known and  $f$  is invertible, the **analytic expression** of  $f^{-1}$  is found by trying to solve for  $x$  the equation:

$f(x) = y$  that is, find the input  $x$  that produces the output  $y$

- ▶ Note that it is not always possible to find it, even in cases where the inverse function exists!

# Inverse of an affine function

Given  $a, b \in \mathbb{R}$ ,  $a \neq 0$ , we have:

$$f : \begin{cases} y = ax + b \\ x \in \mathbb{R} \end{cases}$$

$$f^{-1} : \begin{cases} x = \frac{y-b}{a} \\ y \in \mathbb{R} \end{cases}$$

## Example (1/3)

The function  $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto 2x + 3$  is strictly increasing, hence invertible on  $\mathbb{R}$ , and the equation

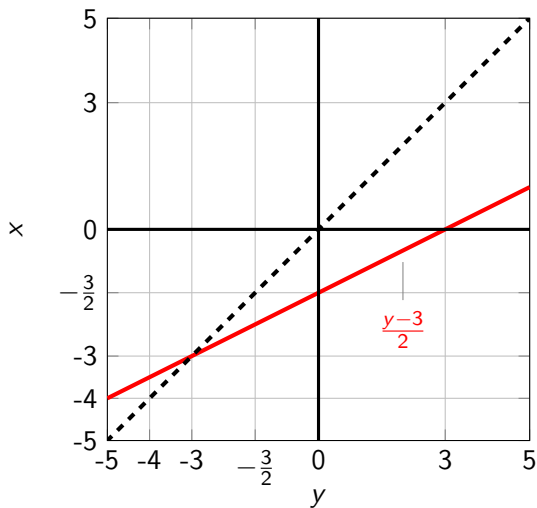
$$2x + 3 = y \text{ solved for } x, \text{ that is } x = \frac{y - 3}{2}$$

gives for  $f^{-1}$  the analytic expression:

$$f^{-1}(y) = \frac{y - 3}{2}$$

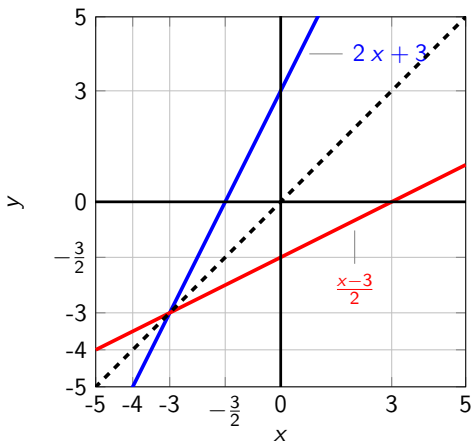
that is, the function:  $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}, y \mapsto \frac{y-3}{2}$ .

## Example (2/3)



## Example (3/3)

The graphs  $y = 2x + 3$  and  $y = \frac{x-3}{2}$  are symmetric with respect to the bisector  $y = x$



# Inverse of $x^2$

We have:

$$f : \begin{cases} y = x^2 \\ x \geq 0 \end{cases}$$

$$f^{-1} : \begin{cases} x = \sqrt{y} \\ y \geq 0 \end{cases}$$

## Example (1/2)

The function  $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto x^2$  is strictly increasing on the interval  $[0, +\infty)$ , hence invertible there; the equation

$$x^2 = y \text{ solved for } x, \text{ that is } x = \sqrt{y}$$

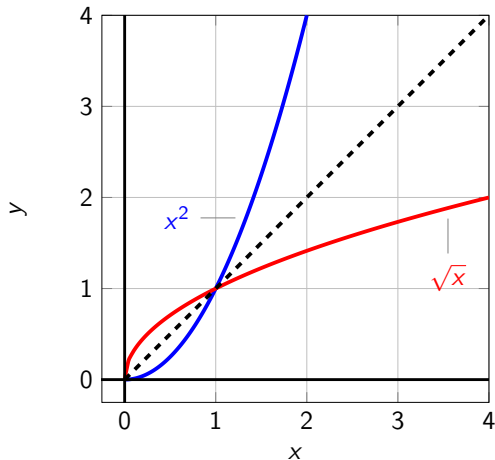
gives for  $f^{-1}$  the analytic expression

$$f^{-1}(y) = \sqrt{y}$$

that is, the function:  $f^{-1} : [0, +\infty) \rightarrow \mathbb{R}, y \mapsto \sqrt{y}$ .

## Example (2/2)

The graphs  $y = x^2$  and  $y = \sqrt{x}$  are symmetric with respect to the bisector  $y = x$



# Inverse of the exponential

Given  $a \in \mathbb{R}_+$ ,  $a \neq 1$ , we have:

$$f : \begin{cases} y = a^x \\ x \in \mathbb{R} \end{cases}$$

$$f^{-1} : \begin{cases} x = \log_a y \\ y > 0 \end{cases}$$

## Example (1/2)

The function  $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto e^x$  is strictly increasing on all of  $\mathbb{R}$  and hence invertible; the equation

$$e^x = y \text{ solved for } x, \text{ that is } x = \log y$$

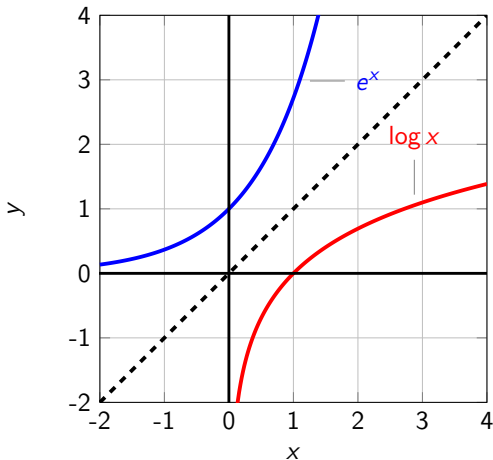
gives for  $f^{-1}$  the analytic expression

$$f^{-1}(y) = \log y$$

that is, the function:  $f^{-1} : (0, +\infty) \rightarrow \mathbb{R}, y \mapsto \log y$ .

## Example (2/2)

The graphs  $y = e^x$  and  $y = \log x$  are symmetric with respect to the bisector  $y = x$



# An inverse that cannot be written explicitly

- ▶ Now consider the function:

$$f(x) = x + e^x$$

Being the sum of two functions that are strictly increasing on all of  $\mathbb{R}$ ,  $f(x)$  is strictly increasing and hence invertible on all of  $\mathbb{R}$  (see the Theorem on strictly monotone functions).

- ▶ However, we would try in vain to solve the equation  $x + e^x = y$  for  $x$ . In other words,  $f^{-1}$  exists, but we cannot write it explicitly.

# Summary

## Graph and expression of the inverse:

- ▶  $(x_0, y_0)$  lies on the graph of  $f$  if and only if  $(y_0, x_0)$  lies on the graph of  $f^{-1}$ : the graphs are symmetric with respect to the bisector  $y = x$
- ▶ The expression of  $f^{-1}$  is found by solving  $f(x) = y$  for  $x$

## Examples:

- ▶  $y = ax + b \Leftrightarrow x = \frac{y-b}{a}$ ;  $y = x^2 (x \geq 0) \Leftrightarrow x = \sqrt{y}$ ;  
 $y = a^x \Leftrightarrow x = \log_a y$
- ▶  $f(x) = x + e^x$  is invertible, but  $f^{-1}$  cannot be written explicitly

# Inverse power functions



# Introduction: Inverse power functions

- ▶ In this section we study the inverse of the **power** functions  $y = x^\alpha$
- ▶ The inverse of a power is again a power, with exponent  $\frac{1}{\alpha}$  (the **roots**)
- ▶ The domain on which the power is invertible depends on the exponent: even powers must be restricted to  $x \geq 0$ , odd powers are invertible on all of  $\mathbb{R}$

# Powers and their inverses (1/2)

Given  $\alpha \in \mathbb{R}, \alpha \neq 0$ , we have:

$$f : \begin{cases} y = x^\alpha \\ x > 0 \end{cases} \qquad f^{-1} : \begin{cases} x = y^{\frac{1}{\alpha}} \\ y > 0 \end{cases}$$

If  $\alpha > 0$  we have:

$$f : \begin{cases} y = x^\alpha \\ x \geq 0 \end{cases} \qquad f^{-1} : \begin{cases} x = y^{\frac{1}{\alpha}} \\ y \geq 0 \end{cases}$$

## Powers and their inverses (2/2)

If  $\alpha = \frac{m}{n} \in \mathbb{Q}$ , with  $m \in \mathbb{Z}$ ,  $n \in \mathbb{N}_+$  odd and coprime, we have:

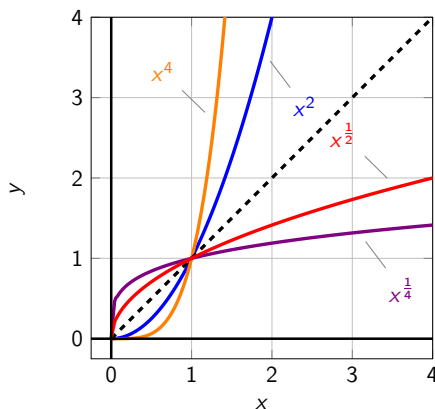
$$f : \begin{cases} y = x^{\frac{m}{n}} \\ x \in \mathbb{R} \end{cases} \qquad f^{-1} : \begin{cases} x = y^{\frac{n}{m}} \\ y \in \mathbb{R} \end{cases}$$

## Even powers

The **even powers**:

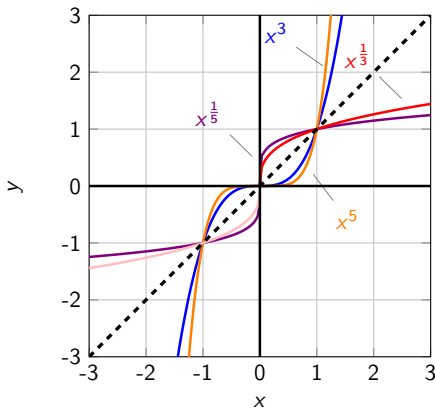
$$x^{2n} \quad \text{with } n = 1, 2, \dots$$

are not invertible on the whole real line, but only on the half-line  $x \geq 0$ .



## Odd powers

The **odd powers**  $x^{2n+1}$  (with  $n = 1, 2, \dots$ ) and the **powers with rational exponent**  $x^{\frac{m}{n}}$  (with  $n, m$  positive odd integers), being strictly increasing monotone functions, are invertible from  $-\infty$  to  $+\infty$ .



# Summary

## Inverse powers:

- ▶ The inverse of  $y = x^\alpha$  ( $x > 0$ ) is  $x = y^{\frac{1}{\alpha}}$ ; if  $\alpha > 0$  this also holds on  $[0, +\infty)$
- ▶ Even powers  $x^{2n}$ : invertible only on  $x \geq 0$  (inverses:  $x^{\frac{1}{2}}, x^{\frac{1}{4}}, \dots$ )
- ▶ Odd powers  $x^{2n+1}$  and rational exponents with  $m, n$  odd: strictly increasing, invertible on all of  $\mathbb{R}$  (inverses:  $x^{\frac{1}{3}}, x^{\frac{1}{5}}, \dots$ )

# Inverse trigonometric functions

# Introduction: Inverse trigonometric functions

- ▶ The trigonometric functions are periodic and hence **not invertible**: to invert them we restrict them to intervals on which they are strictly monotone
- ▶ In this way we obtain the **arcsine**, the **arccosine** and the **arctangent**, with their graphs
- ▶ These functions are used to write the solutions of trigonometric equations and inequalities with non-standard angles

# Trigonometric functions are not invertible

- ▶ Being periodic, the trigonometric functions cannot be invertible. Indeed, for example, the equation

$$y = \sin x$$

has infinitely many solutions if  $-1 \leq y \leq 1$  (the output  $y$  corresponds to infinitely many inputs), or has no real solutions if  $|y| > 1$ .

- ▶ To speak of inverse functions of sine, cosine and tangent, we will need to restrict ourselves to intervals on which these functions are strictly monotone and therefore invertible.

# Arcsine

- An interval on which the sine function is invertible is  $[-\frac{\pi}{2}, \frac{\pi}{2}]$ .

## Definition 3: of arcsine

*The inverse function of the sine on the interval  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  is the **arcsine**:*

$$f : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right], \quad f : y \mapsto \arcsin y$$

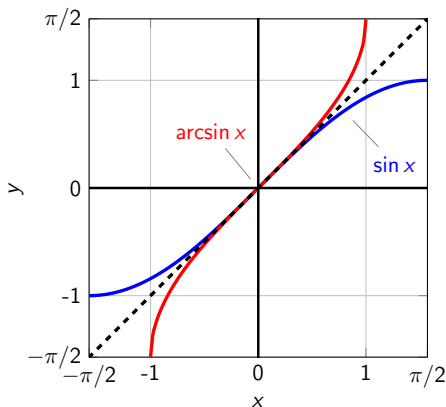
We have:

$$f : \begin{cases} y = \sin x \\ x \in [-\frac{\pi}{2}, \frac{\pi}{2}] \end{cases}$$

$$f^{-1} : \begin{cases} x = \arcsin y \\ y \in [-1, 1] \end{cases}$$

# Graph of the arcsine

The graph of the arcsine is obtained from the arc of the sine curve restricted to the interval  $[-\frac{\pi}{2}, \frac{\pi}{2}]$  (strictly monotone), by symmetry with respect to the bisector  $y = x$ :



# Arccosine

- An interval on which the cosine function is invertible is  $[0, \pi]$ .

## Definition 4: of arccosine

*The inverse function of the cosine on the interval  $[0, \pi]$  is the **arccosine**:*

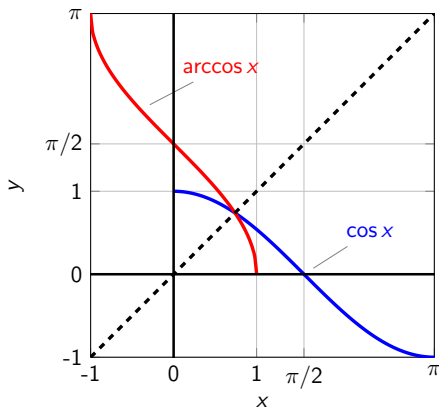
$$f : [-1, 1] \rightarrow [0, \pi], \quad f : y \mapsto \arccos y$$

We have:

$$f : \begin{cases} y = \cos x \\ x \in [0, \pi] \end{cases} \qquad f^{-1} : \begin{cases} x = \arccos y \\ y \in [-1, 1] \end{cases}$$

# Graph of the arccosine

The graph of the arccosine is obtained from the arc of the cosine curve restricted to the interval  $[0, \pi]$  (strictly monotone), by symmetry with respect to the bisector  $y = x$ :



# Arctangent

- Note that on the interval  $(-\frac{\pi}{2}, \frac{\pi}{2})$  the tangent is strictly monotone and hence invertible. Its inverse function is called the arctangent ( $\arctan$ ), and it is defined on  $\mathbb{R}$ .

## Definition 5: of arctangent

*The inverse function of the tangent on the interval  $(-\frac{\pi}{2}, \frac{\pi}{2})$  is the **arctangent**:*

$$f : \mathbb{R} \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right), \quad f : y \mapsto \arctan y$$

# Arctangent

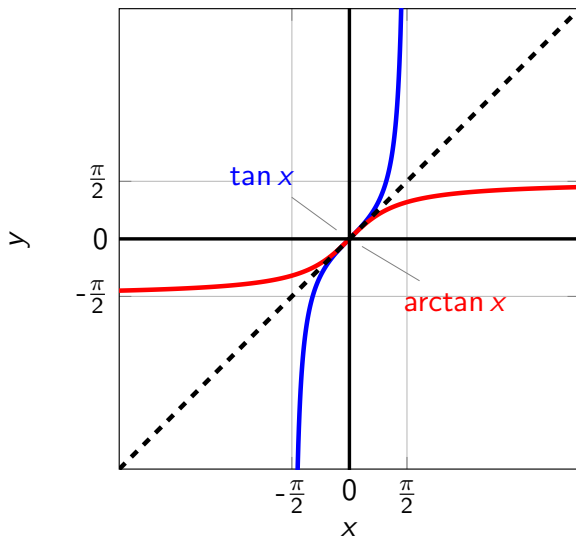
We have:

$$f : \begin{cases} y = \tan x \\ x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \end{cases}$$

$$f^{-1} : \begin{cases} x = \arctan y \\ y \in \mathbb{R} \end{cases}$$

- ▶ The graph of the arctangent is obtained from that of the tangent restricted to the interval  $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$  (strictly monotone), by symmetry with respect to the bisector  $y = x$ .

# Graph of the arctangent



# Trigonometric equations and inequalities

- ▶ By means of the inverse trigonometric functions, we can express the solutions of a **trigonometric equation or inequality** when it involves angles that are not standard angles.

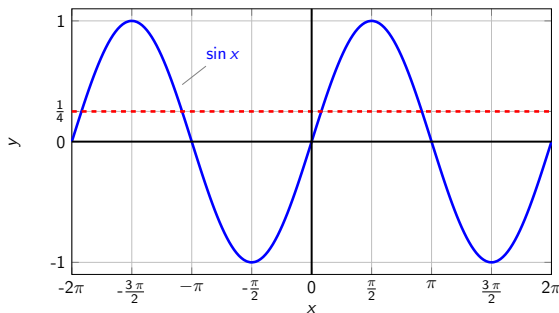
## Example (1/3)

The solutions of:

$$\sin x = \frac{1}{4}$$

are:

$$(i) \ x = \arcsin \frac{1}{4} + 2k\pi; \quad (ii) \ x = \pi - \arcsin \frac{1}{4} + 2k\pi \quad (k \in \mathbb{Z})$$



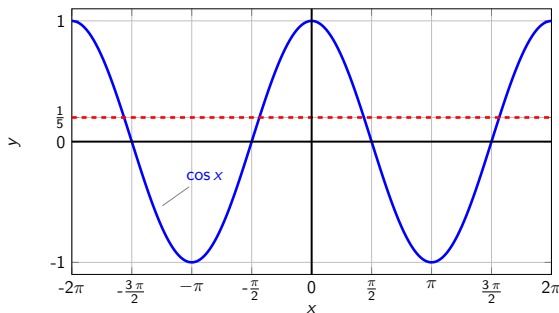
## Example (2/3)

The solutions of:

$$\cos x < \frac{1}{5}$$

are:

$$\arccos \frac{1}{5} + 2k\pi < x < 2\pi - \arccos \frac{1}{5} + 2k\pi \quad (k \in \mathbb{Z})$$



## Example (3/3)

The solutions of:

$$\tan x \geq 3$$

are:

$$\arctan 3 + k \pi \leq x < \frac{\pi}{2} + k \pi \quad (k \in \mathbb{Z})$$

# Summary

## Inverse trigonometric functions:

- ▶  $\arcsin : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ , inverse of the sine restricted to  $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- ▶  $\arccos : [-1, 1] \rightarrow [0, \pi]$ , inverse of the cosine restricted to  $[0, \pi]$
- ▶  $\arctan : \mathbb{R} \rightarrow (-\frac{\pi}{2}, \frac{\pi}{2})$ , inverse of the tangent restricted to  $(-\frac{\pi}{2}, \frac{\pi}{2})$

## Application:

- ▶ They allow us to write the solutions of trigonometric equations and inequalities with non-standard angles

# Inverse hyperbolic functions

# Introduction: Inverse hyperbolic functions

- ▶ In this section we invert the **hyperbolic sine** and the **hyperbolic cosine** (the latter restricted to  $x \geq 0$ )
- ▶ Unlike the trigonometric functions, here we can find the **analytic expression** of the inverse, by solving a quadratic equation in  $e^x$
- ▶ We obtain the **inverse hyperbolic sine** and the **inverse hyperbolic cosine**, expressed through the logarithm

## Inverse hyperbolic sine (1/3)

- Consider the **hyperbolic sine** function:

$$y = \operatorname{Sh} x = \frac{e^x - e^{-x}}{2}$$

It is defined and strictly increasing on all of  $\mathbb{R}$ , therefore it is invertible.

- To solve the equation for  $x$ , multiplying both sides by  $e^x$  we obtain:

$$\begin{aligned} 0 &= e^x y - e^x \frac{e^x - e^{-x}}{2} \\ &= 2 e^x y - (e^x e^x - \underbrace{e^x e^{-x}}_{=e^{x-x}=e^0}) \\ &= 2 y e^x - e^{2x} + 1 \\ &= e^{2x} - 2 y e^x - 1 \end{aligned}$$

## Inverse hyperbolic sine (2/3)

- ▶ This is a quadratic equation in the unknown  $e^x$ . Setting  $t = e^x$  we obtain

$$0 = t^2 - 2 y t - 1$$

We obtain:

$$t = y \pm \sqrt{y^2 + 1} \quad \text{and substituting back} \quad e^x = y \pm \sqrt{y^2 + 1}$$

since  $e^x > 0$ , the solution with the minus sign must be discarded. Hence we are left with:

$$e^x = \underbrace{y + \sqrt{y^2 + 1}}_{>0, \forall y \in \mathbb{R}} \quad \text{and hence} \quad x = \log \left( y + \sqrt{y^2 + 1} \right)$$

## Inverse hyperbolic sine (3/3)

- This is the analytic expression of the inverse function of  $\text{Sh } x$ , which is called the **inverse hyperbolic sine** (area hyperbolic sine), and is also denoted by  $\text{SettSh}$ . It is defined for every real  $y$ .

### Definition 6: of inverse hyperbolic sine

*The inverse function of the hyperbolic sine is the **inverse hyperbolic sine**:*

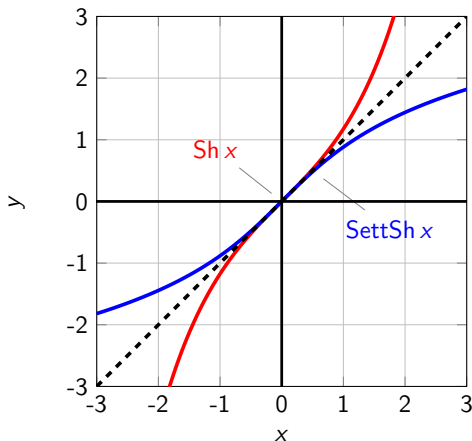
$$f : \mathbb{R} \rightarrow \mathbb{R}, \quad f : y \mapsto \text{SettSh } y$$

We have:

$$f : \begin{cases} y = \text{Sh } x \\ x \in \mathbb{R} \end{cases} \qquad f^{-1} : \begin{cases} x = \text{SettSh } y \\ y \in \mathbb{R} \end{cases}$$

## Graph of the inverse hyperbolic sine

The graph of the inverse hyperbolic sine is obtained from that of the hyperbolic sine on  $\mathbb{R}$  (strictly monotone), by symmetry with respect to the bisector  $y = x$ :



## Inverse hyperbolic cosine (1/3)

- Consider the **hyperbolic cosine** function:

$$y = \operatorname{Ch} x = \frac{e^x + e^{-x}}{2}$$

It is defined on all of  $\mathbb{R}$ , strictly increasing for  $x \geq 0$ , decreasing for  $x \leq 0$ . Therefore it is not invertible on all of  $\mathbb{R}$ .

- Its restriction to  $x \geq 0$ , however, is invertible. We want to determine the analytic expression of the inverse function of this restriction.
- Proceeding as before, we obtain:

$$e^x = y \pm \sqrt{y^2 - 1}$$

## Inverse hyperbolic cosine (2/3)

- ▶ This time both numbers  $y \pm \sqrt{y^2 - 1}$  are positive; recall, however, that we are reasoning only for  $x \geq 0$ , which is equivalent to choosing the plus sign. Therefore:

$$x = \log \left( y + \sqrt{y^2 - 1} \right)$$

This is the analytic expression of the inverse function of  $\text{Ch } x$ , which is called the inverse hyperbolic cosine (area hyperbolic cosine), and is also denoted by  $\text{SettCh}$ . Note that it is defined for  $y \geq 1$ .

## Inverse hyperbolic cosine (3/3)

### Definition 7: of inverse hyperbolic cosine

*The inverse function of the hyperbolic cosine on the interval  $[0, +\infty)$  is the **inverse hyperbolic cosine**:*

$$f : [1, +\infty) \rightarrow [0, +\infty), \quad f : y \mapsto \operatorname{SettCh} y$$

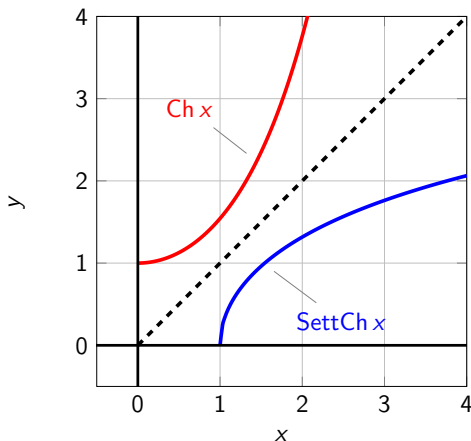
We have:

$$f : \begin{cases} y = \operatorname{Ch} x \\ x \in [0, +\infty) \end{cases}$$

$$f^{-1} : \begin{cases} x = \operatorname{SettCh} y \\ y \in [1, +\infty) \end{cases}$$

## Graph of the inverse hyperbolic cosine

The graph of the inverse hyperbolic cosine is obtained from that of the hyperbolic cosine on  $[0, +\infty)$  (strictly monotone), by symmetry with respect to the bisector  $y = x$ .



## Example (1/2)

- ▶ The equation:

$$\operatorname{Sh} x = 2$$

has the unique solution:

$$x = \operatorname{SettSh} 2 = \log(2 + \sqrt{5})$$

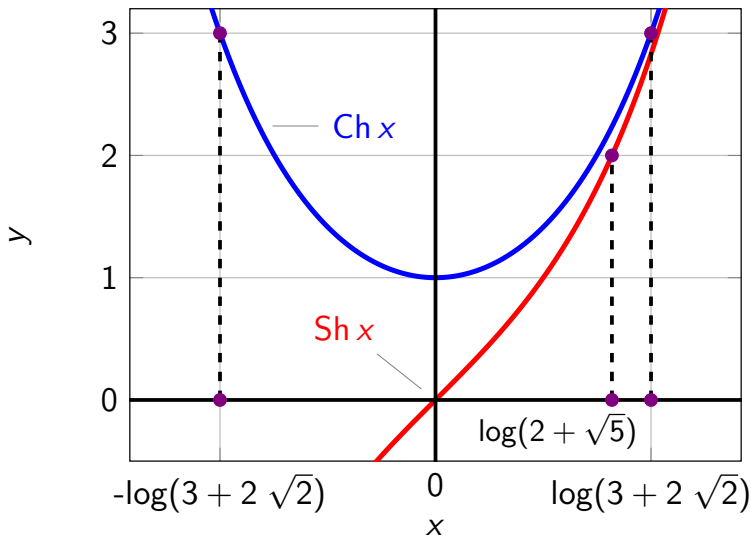
- ▶ The equation:

$$\operatorname{Ch} x = 3$$

has two solutions:

$$x = \pm \operatorname{SettCh} 3 = \pm \log(3 + 2\sqrt{2})$$

## Example (2/2)



## Inverse hyperbolic functions:

- ▶ Sh is strictly increasing on  $\mathbb{R}$ : its inverse is the inverse hyperbolic sine  $\text{SettSh } y = \log \left( y + \sqrt{y^2 + 1} \right)$ , defined for every  $y \in \mathbb{R}$
- ▶ Ch is invertible only if restricted to  $x \geq 0$ : its inverse is the inverse hyperbolic cosine  $\text{SettCh } y = \log \left( y + \sqrt{y^2 - 1} \right)$ , defined for  $y \geq 1$
- ▶ The expressions are obtained by solving a quadratic equation in  $t = e^x$

# Summary (1/2)

## Invertible functions:

- ▶  $f$  is invertible if it is injective; the inverse  $f^{-1}$  is defined on  $f(D)$
- ▶ Strictly monotone  $\Rightarrow$  invertible; even or periodic  $\Rightarrow$  not invertible
- ▶ The graph of  $f^{-1}$  is symmetric to that of  $f$  with respect to the bisector  $y = x$
- ▶ The expression of  $f^{-1}$  is obtained by solving  $f(x) = y$  for  $x$  (not always possible)

## Summary (2/2)

### Inverses of the elementary functions:

- ▶ Powers: the inverse of  $x^\alpha$  is  $y^{\frac{1}{\alpha}}$  (even powers only for  $x \geq 0$ )
- ▶ Exponential and logarithm:  $y = a^x \leftrightarrow x = \log_a y$
- ▶ Trigonometric (on intervals of monotonicity): arcsin, arccos, arctan
- ▶ Hyperbolic:  $\text{SettSh } y = \log \left( y + \sqrt{y^2 + 1} \right),$   
 $\text{SettCh } y = \log \left( y + \sqrt{y^2 - 1} \right)$